Parallelization of Algorithms - Final Project Guide (2-3 students max in a group, you can do it solo if you want to)

Project Guidelines

Objective: Implement a parallel algorithm using shared-memory programming, MPI, explicit data ownership, correctness tests, and measured performance analysis. You are expected to analyze performance, demonstrate scalability, and present findings clearly.

Deliverables:

- Live demonstration of the working project.
- A presentation including:
  - Problem definition and motivation.
  - Description of sequential and parallel implementations.
  - Speedup, efficiency, and scalability graphs.
  - Challenges and optimization strategies.
- Submitted code with comments and execution instructions, presentation slides via Moodle.

Project Ideas

Project Title	**Recommended Team Size**
Game of Life Simulation	2 students
Lid-Driven Cavity Fluid Simulation	3 students
Hybrid Merge Sort	2 students
Parallel Image Filtering	2 students
Parallel Word Count	2 students
BFS/Dijkstra on Large Graphs	2–3 students
Log File Analyzer	2 students
Sparse Matrix-Vector Multiplication	3 students
3D Heat Diffusion	3 students
Parallel File Compression Utility	3 students
Ray Tracing Engine (CPU-only)	3 students
Epidemic Spread Simulation	3 students
Traffic Flow Simulation	3 students
YOUR OWN PROJECT	Propose/Discuss with me first! Otherwise, will not be counted!

Grading Criteria (Total: 100 Points)

Category	Points	Description
Problem Selection & Scope	5	Relevance, challenge level, and use of MPI/OpenMP or both
Correctness	20	Correct implementation of serial and parallel versions
Use of OpenMP/MPI or both	25	Effective and appropriate use of both technologies
Performance Analysis	20	Speedup, efficiency, scalability with proper evaluation
Code Quality	10	Clean, readable, and documented code
Presentation Quality	10	Clear explanation, graphs, and structured content
Originality & Insight	10	Creative solution, hybrid technique, or optimization insight

Plagiarism and Reproducibility

All code must be original and explained during the live demo. Projects will be tested for reproducibility. Students must be able to explain every part of their code. External libraries or code snippets must be properly cited. Projects suspected of plagiarism will receive a failing grade.